

Target Detection and Localization Using MIMO Radars and Sonars

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Abstract—In this paper, we propose a new space-time coding configuration for target detection and localization by radar or sonar systems. In common active array systems, the transmitted signal is usually coherent between the different elements of the array. This configuration does not allow array processing in the transmit mode. However, space-time coding of the transmitted signals allows to digitally steer the beam pattern in the transmit in addition to the received signal. The ability to steer the transmitted beam pattern, helps to avoid beam shape loss. We show that the configuration with spatially orthogonal signal transmission is equivalent to additional virtual sensors which extend the array aperture with virtual spatial tapering. These virtual sensors can be used to form narrower beams with lower sidelobes and, therefore, provide higher performance in target detection, angular estimation accuracy, and angular resolution. The generalized likelihood ratio test for target detection and the maximum likelihood and Cramér–Rao bound for target direction estimation are derived for an arbitrary signal coherence matrix. It is shown that the optimal performance is achieved for orthogonal transmitted signals. Target detection and localization performances are evaluated and studied theoretically and via simulations.

Index Terms—Cramér–Rao bound (CRB), generalized likelihood ratio test (GLRT), maximum likelihood, MIMO radars, MIMO sonars, orthogonal signal transmission, space–time coding, transmit beamforming, virtual sensors.

I. INTRODUCTION

ACTIVE target detection and localization systems, such as radars or active sonars, usually transmit a directional beam, and the target echo signal is processed in the receive mode. In the last two decades, array processing of the received signal has been intensively investigated (see, for example, [1]). However, this configuration does not allow array processing in the transmit mode. In fact, the transmission is usually performed using the phased array technique or other beam steering methods. Array processing in both transmit and receive modes is possible when the transmitted signals are spatially coded, i.e., spatially orthogonal signals. This paper addresses the problem of target detection and localization by active array using spatially coded signals.

Transmission of orthogonal signals from an array is commonly used in communication systems [2]. Passive localization of orthogonal signals with known waveforms was investigated

in [3]. In [4], it is shown that the conventional configuration of one transmitter and two receivers and an alternative configuration of two transmitters and one receiver are equivalent in terms of Cramér–Rao bound (CRB) on bearing estimation. This configuration requires radiating two orthogonal signals from two transmitters. The potential advantage of this configuration over the conventional one is in applications where the receiving elements are to be placed on a platform of limited size. The results in [4] were extended in [5], in which three possible combinations of four transmitters/receivers were investigated: 1) one transmitter and three receivers, 2) two transmitters and two receivers, and 3) three transmitters and one receiver. It was found that these configurations have identical performance in terms of angle estimation accuracy, where the transmitting signals are orthogonal. In [6], spatio-temporal coding for an antenna array was introduced, and it was shown that a single receiver is sufficient for digital beamforming. Fishler *et al.* [7] also investigated the problem of orthogonal signal transmission for multiple-input multiple-output (MIMO) radar. They assumed a multistatic radar in which the spacing between the elements of the array is very large, and that the transmitter and the receiver of the radar are separated such that they experience an angular spread.

In [8], a novel configuration for array processing using space-time coding of the transmitted signal was presented. This configuration does not assume a multistatic radar, with an arbitrary signal coherence across the radar elements.

In this paper, we analyze the properties of the space-time coding configuration for target detection and localization. In particular, the advantages of the proposed configuration are analytically demonstrated and compared to the conventional coherent signal transmission case. The main advantages of this new configuration are as follows:

- digital beamforming of the transmitted beams in addition to the received beams, therefore avoiding beam shape loss in cases when the target is not in the center of the beam;
- extension of the array aperture by virtual sensors, therefore obtaining narrower beams;
- virtual spatial tapering of the extended array aperture, therefore obtaining lower side lobes;
- improving the angular resolution by using the information in the transmit and the receive modes;
- increasing the upper limit on the number of targets which can be detected and localized by the array (this is attributed to the virtual sensors);
- decreasing the spatial transmitted peak power density.

This paper is organized as follows. The spatially coded signal model is presented in Section II. In Section III, the sufficient statistic (SS) for detection and estimation algorithms are derived. The model's properties are analyzed in Section IV. The

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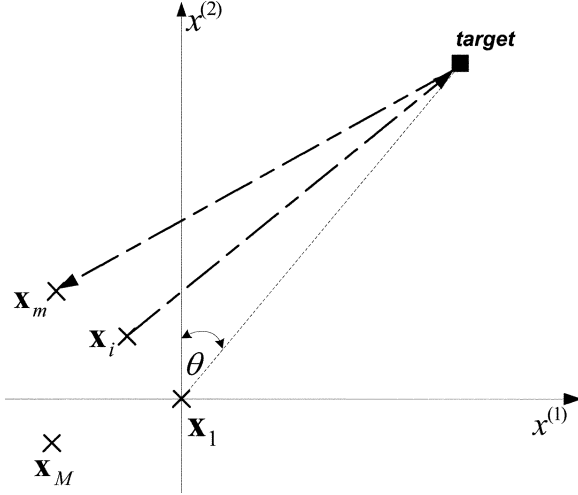


Fig. 1. Array configuration.

maximum likelihood (ML) estimator, the CRB for target localization, and the generalized likelihood ratio test (GLRT) for target detection are derived in Section V. The proposed concept is tested via a few examples and simulations, which appear in Section VI. The main results of this paper are discussed and concluded in Section VII.

II. SPATIALLY CODED SIGNAL MODEL

Consider an M element antenna array transmitting M narrow-band signals. The samples of baseband equivalent signals are denoted by the vectors $\{\mathbf{s}[n]\}_{n=1}^N$ with coherence matrix

$$\mathbf{R}_s = \frac{1}{N} \sum_{n=1}^N \mathbf{s}[n] \mathbf{s}^H[n] = \begin{bmatrix} 1 & \beta_{12} & \cdots & \beta_{1M} \\ \beta_{21} & 1 & \cdots & \beta_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{M1} & \beta_{M2} & \cdots & 1 \end{bmatrix} \quad (1)$$

where n represents the time index, β_{ij} is the complex correlation coefficient between the i th and j th signals, and H denotes the Hermitian operation. The phases of $\{\beta_{ij}\}_{i,j=1}^M$ control the transmitted beam direction of the coherent component. In the case of orthonormal transmitted signals ($\{\beta_{ij}\}_{i \neq j} = 0$) and the coherence matrix is an identity matrix: $\mathbf{R}_s = \mathbf{I}_M$, i.e., omnidirectional transmission. In common radar systems, coherent signals are transmitted by the array and therefore the rank of $\mathbf{R}_s$ is equal to one.¹

Let $\mathbf{x}_m \triangleq [x_m^{(1)}, x_m^{(2)}]^T$ denote the location of the m th element of the array with T denoting the transposition operation (see Fig. 1). In the presence of a single target at direction θ in a multipath-free environment, the received signal by the m th element is given by

$$y_m[n] = \alpha \sum_{i=1}^M A_{im}(\theta) s_i[n] + w_m[n] \quad m=1, \dots, M$$

$$n=1, \dots, N \quad (2)$$

¹The different elements transmit the same signal with phase shifts for beam steering.

where α stands for the complex amplitude of the received signal, $w_m[n]$ is the additive noise at the m th element, and $A_{im}(\theta) = \exp(-j\omega_c \tau_{im}(\theta))$ describes the total phase delay of the signal, transmitted by the i th element and received by the m th element, where ω_c is the carrier frequency. The total delay from the i th transmitting element to the m th receiving element for the far-field case can be written as

$$\omega_c \tau_{im}(\theta) = \omega_c \left(\underbrace{\tau_i(\theta)}_{\text{transmit}} + \underbrace{\tau_m(\theta)}_{\text{receive}} \right) = \mathbf{k}^T(\theta)(\mathbf{x}_i + \mathbf{x}_m) \quad (3)$$

where $\mathbf{k}(\theta) = (2\pi/\lambda)[\sin \theta, \cos \theta]^T$ and λ stands for the signal wavelength.

Let $a_m(\theta)$ denote the response of the m th element. Then, the i th element of the array response $A_{im}(\theta)$ can be decomposed as

$$A_{im}(\theta) = \exp(-j\omega_c (\tau_i(\theta) + \tau_m(\theta)))$$

$$= a_i(\theta) a_m(\theta) \quad i, m = 1, \dots, M. \quad (4)$$

Note that the elements of $\mathbf{A}(\theta)$ depend on θ through all possible combinations of delays in transmit and receive modes. In fact, $A_{im}(\theta)$ is the array response for transmit from the i th element and receive by the m th element. Hence, the array response matrix can be defined as

$$\mathbf{A}(\theta) = \mathbf{a}(\theta) \mathbf{a}^T(\theta) = [\mathbf{A}_{:1}(\theta), \dots, \mathbf{A}_{:M}(\theta)] \quad (5)$$

in which $\mathbf{a}(\theta)$ is the transmitted or the received array response vector. In matrix notation, (2) can be written as

$$\mathbf{y}[n] = \alpha \mathbf{A}(\theta) \mathbf{s}[n] + \mathbf{w}[n] \quad n = 1, \dots, N \quad (6)$$

where $\mathbf{y}[n]$, $\mathbf{s}[n]$, and $\mathbf{w}[n]$ are vectors of the received signal, the transmitted signal, and the additive noise, respectively.

In sensors with range or Doppler estimation capability, the model should include also the target range and Doppler. Typically, in these sensors, the ML estimator of the target direction, range, and Doppler is implemented by processing the receiving channels over time, obtaining multichannel measurements for each considered range-Doppler bin. The above model refers to a single range-Doppler bin. Specifically, the target detection and localization algorithms presented in this paper should be applied for each range-Doppler bin individually. In the above model, multiple targets can be allowed if they do not share the same range-Doppler bin. In the following, this model is extended to allow multiple targets in the considered range-Doppler bin.

In the case of L targets in the given range-Doppler bin, (6) is modified to

$$\mathbf{y}[n] = \sum_{l=1}^L \alpha_l \mathbf{A}(\theta_l) \mathbf{s}[n] + \mathbf{w}[n] \quad n = 1, \dots, N. \quad (7)$$

Let $\boldsymbol{\xi} = [\boldsymbol{\theta}^T \tilde{\boldsymbol{\alpha}}^T]^T$ denote the vector of unknown parameters, which includes the directions of arrival (DOAs) and the complex amplitudes of the L targets, where $\boldsymbol{\theta} = [\theta_1, \dots, \theta_L]^T$ and $\tilde{\boldsymbol{\alpha}} = [\tilde{\boldsymbol{\alpha}}_1^T, \dots, \tilde{\boldsymbol{\alpha}}_L^T]^T = [\text{Re}\{\alpha_1\}, \text{Im}\{\alpha_1\}, \dots, \text{Re}\{\alpha_L\}, \text{Im}\{\alpha_L\}]^T$. The vectors $\boldsymbol{\theta}$ and $\tilde{\boldsymbol{\alpha}}$ are considered as deterministic unknown.

The noise vectors $\{\mathbf{w}[n]\}_{n=1}^N$ are assumed to be independent, zero-mean complex Gaussian with known covariance matrix $\mathbf{R}_w$. With no loss of generality, we can assume that $\mathbf{R}_w = \sigma_w^2 \mathbf{I}_M$, where $\mathbf{I}_M$ is an identity matrix of size M . If this assumption is not satisfied, the model in (7) can be prewhitened. In many practical scenarios, in the presence of clutter or jammer, the noise covariance matrix is unknown. In these cases, the noise can be treated as a composition of additional interference sources. This problem can be solved by either multiple target localization techniques, as is modeled below, or by adaptive methods such as the Capon's beamformer.

III. SUFFICIENT STATISTIC FOR DOA ESTIMATION

According to the assumptions stated in the previous section, the measurement vectors are independent complex Gaussian vectors with $\mathbf{y}[n] \sim N^c(\sum_{l=1}^L \alpha_l \mathbf{A}(\theta_l) \mathbf{s}[n], \sigma_w^2 \mathbf{I}_M)$, where N^c denotes the complex Gaussian distribution. The log-likelihood function for estimating $\boldsymbol{\xi}$ from the data $\mathbf{Y} = (\mathbf{y}[1], \dots, \mathbf{y}[N])$ is derived in Appendix A and given by

$$\begin{aligned} \log f_{\mathbf{Y}}(\mathbf{Y}; \boldsymbol{\xi}) &= -MN \log(\pi \sigma_w^2) - \frac{1}{\sigma_w^2} \sum_{n=1}^N \|\mathbf{y}[n]\|^2 \\ &+ \frac{2\sqrt{N}}{\sigma_w^2} \operatorname{Re} \left\{ \sum_{l=1}^L \alpha_l^* \sum_{m=1}^M \mathbf{A}_{:,m}^H(\theta_l) \boldsymbol{\eta}_m(\mathbf{Y}) \right\} \\ &- \frac{N}{\sigma_w^2} \sum_{k,l=1}^L \alpha_k^* \alpha_l \mathbf{a}^H(\theta_l) \mathbf{R}_s^H \mathbf{a}(\theta_k) \mathbf{a}^H(\theta_k) \mathbf{a}(\theta_l) \end{aligned} \quad (8)$$

where $\mathbf{A}_{:,m}(\theta)$ is the m th column of $\mathbf{A}(\theta)$, $*$ denotes the conjugate operation, and $\boldsymbol{\eta}_m$ is the m th sufficient statistic, defined as

$$\boldsymbol{\eta}_m \triangleq \frac{1}{\sqrt{N}} \sum_{n=1}^N \mathbf{y}[n] s_m^*[n] \quad m = 1, \dots, M \quad (9)$$

which is obtained by matching the observed data to the m th signal $\{s_m[n]\}_{n=1}^N$. Moreover, the sufficient statistic matrix can be defined as

$$\mathbf{E} = [\boldsymbol{\eta}_1, \dots, \boldsymbol{\eta}_M] = \frac{1}{\sqrt{N}} \sum_{n=1}^N \mathbf{y}[n] \mathbf{s}^H[n]. \quad (10)$$

It can be shown that for nonorthogonal signals, the sufficient statistic $\{\boldsymbol{\eta}_m\}_{m=1}^M$ are statistically dependent. For simplicity of the algorithms, we are interested in independent sufficient statistic $\{\tilde{\boldsymbol{\eta}}_m\}_{m=1}^M$, which can be obtained as follows. We will first assume that the matrix $\mathbf{R}_s$ is nonsingular, and then we will refer to the general case. The matrix $\mathbf{R}_s$ from (1) can be decomposed using singular value decomposition (SVD) as $\mathbf{R}_s = \mathbf{U} \boldsymbol{\Lambda} \mathbf{U}^H$, where $\mathbf{U}$ and $\boldsymbol{\Lambda}$ are the matrices of eigenvectors and eigenvalues of $\mathbf{R}_s$, respectively. Accordingly, $\mathbf{s}[n]$ is a linear transformation of a vector of independent signals defined as

$$\tilde{\mathbf{s}}[n] \triangleq \boldsymbol{\Lambda}^{-1/2} \mathbf{U}^H \mathbf{s}[n]. \quad (11)$$

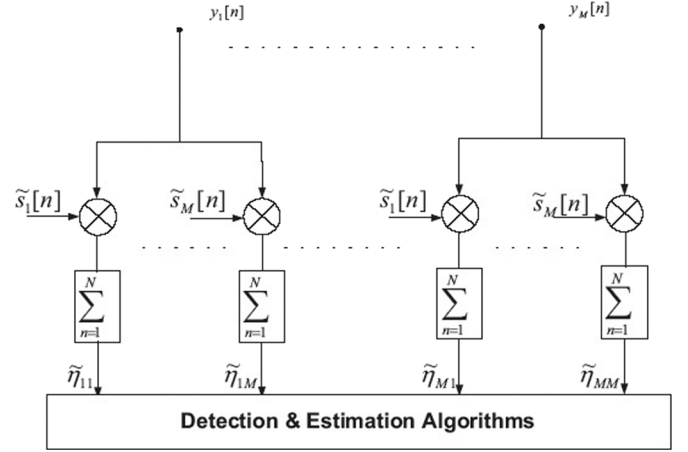


Fig. 2. Sufficient statistic extraction.

The independent sufficient statistic vector can be obtained as

$$\tilde{\boldsymbol{\eta}} \triangleq \operatorname{vec}[\tilde{\boldsymbol{\eta}}_1, \dots, \tilde{\boldsymbol{\eta}}_M] = \operatorname{vec} \left(\frac{1}{\sqrt{N}} \sum_{n=1}^N \mathbf{y}[n] \tilde{\mathbf{s}}^H[n] \right). \quad (12)$$

The configuration for obtaining the sufficient statistic from the data is described in Fig. 2. Actually, the sufficient statistic can be obtained by a *matched filter*: temporal matching the measurement vectors to different signal subspace components $\tilde{\mathbf{s}}[n]$. Insertion of (7) and (11) into (12) yields

$$\begin{aligned} \tilde{\boldsymbol{\eta}} = \operatorname{vec} \left(\sum_{l=1}^L \alpha_l \mathbf{A}(\theta_l) \frac{1}{\sqrt{N}} \sum_{n=1}^N \mathbf{s}[n] \mathbf{s}^H[n] \mathbf{U} \boldsymbol{\Lambda}^{-1/2} \right. \\ \left. + \frac{1}{\sqrt{N}} \sum_{n=1}^N \mathbf{w}[n] \tilde{\mathbf{s}}^H[n] \right). \end{aligned} \quad (13)$$

By recalling the definition of $\mathbf{R}_s$ from (1) and using its SVD decomposition form, (13) can be rewritten as

$$\begin{aligned} \tilde{\boldsymbol{\eta}} = \sum_{l=1}^L \alpha_l \operatorname{vec} \left(\mathbf{A}(\theta_l) \sqrt{N} \mathbf{U} \boldsymbol{\Lambda}^{1/2} \right) \\ + \operatorname{vec} \left(\frac{1}{\sqrt{N}} \sum_{n=1}^N \mathbf{w}[n] \tilde{\mathbf{s}}^H[n] \right). \end{aligned} \quad (14)$$

Finally, (14) can be written in the form

$$\tilde{\boldsymbol{\eta}} = \sum_{l=1}^L \alpha_l \mathbf{d}_{\boldsymbol{\beta}}(\theta_l) + \mathbf{v} \quad (15)$$

$$\mathbf{d}_{\boldsymbol{\beta}}(\theta) \triangleq \sqrt{N} \operatorname{vec} \left(\mathbf{A}(\theta) \mathbf{U} \boldsymbol{\Lambda}^{1/2} \right) \quad (16)$$

where $\mathbf{d}_{\boldsymbol{\beta}}(\theta_l)$ is the equivalent array response of size M^2 at the direction θ_l and $\boldsymbol{\beta} = \operatorname{vec}(\{\beta_{i,j}\}_{i,j=1,j>i}^M)$. The subscript $\boldsymbol{\beta}$ denotes dependence of the new steering vector on the signal correlation coefficients and $\mathbf{v} = (1/\sqrt{N}) \operatorname{vec}(\sum_{n=1}^N \mathbf{w}[n] \tilde{\mathbf{s}}^H[n])$, which is zero-mean complex Gaussian with $\mathbf{v} \sim N^c(\mathbf{0}, \sigma_w^2 \mathbf{I}_{M^2})$. Actually, (15) states an equivalent model to (7).

If the matrix $\mathbf{R}_s$ is singular, then part of its eigenvalues are equal to zero. In such a case, the matrix $\mathbf{A}$ is changed to $\lim_{\epsilon \rightarrow 0}(\mathbf{A} + \epsilon \mathbf{I}_M)$, and the above procedure can be repeated with this modified eigenvalues matrix. The final result in this case is independent of ϵ , and therefore the “lim” can be dropped. Accordingly, (15) and (16) hold also for singular $\mathbf{R}_s$.

IV. MODEL'S PROPERTIES

In this section, the properties of the equivalent model of (15) are investigated and illustrated.

A. Virtual Aperture Extension

Recalling (16), the equivalent array response $\mathbf{d}_\beta(\theta)$ can be calculated for coherent and orthogonal signals. In the coherent signal case, the matrix $\mathbf{R}_s = \mathbf{u}\mathbf{u}^H$ has a single nonzero eigenvalue. Therefore, $\mathbf{U}\mathbf{A}^{1/2} = [\mathbf{u} \mathbf{0} \cdots \mathbf{0}]$ and the equivalent steering vector becomes

$$\mathbf{d}_{\beta\text{coherent}}(\theta) = \sqrt{N}\text{vec}(\mathbf{A}(\theta)[\mathbf{u} \mathbf{0} \cdots \mathbf{0}]) \quad (17)$$

where $\mathbf{0}$ represents a vector of size M , with zero elements. By substitution of (5) into (17), one obtains

$$\mathbf{d}_{\beta\text{coherent}}(\theta) = \sqrt{N}\text{vec}([\mathbf{a}(\theta)(\mathbf{a}^T(\theta)\mathbf{u}) \mathbf{0} \cdots \mathbf{0}]). \quad (18)$$

The equivalent array response is given by the steering vector in the receive mode $\mathbf{a}(\theta)$ multiplied by the gain achieved in the transmit mode $(\mathbf{a}^T(\theta)\mathbf{u}) = \sum_{m=1}^M u_m a_m(\theta)$, where $\mathbf{u} = [u_1, \dots, u_M]^T$ is the weighting vector in the transmit mode. This gain is decreased due to the beam shape loss when the target is not located in the center of the transmit beam. In the particular case of $\mathbf{u} = \mathbf{1}$, with $\mathbf{1}$ denoting a column vector of size M whose elements are equal to one, the transmit beam is directed to the array broadside and $\mathbf{R}_s = \mathbf{1}\mathbf{1}^T$, that is, $\{\beta_{i,j}\}_{i,j=1}^M = 1$. The equivalent array response in this case, denoted by $\mathbf{d}_{\beta=1}(\theta)$, can be written as

$$\mathbf{d}_{\beta=1}(\theta) = \sqrt{N}\text{vec}\left(\left[\mathbf{a}(\theta) \sum_{m=1}^M a_m(\theta) \mathbf{0} \cdots \mathbf{0}\right]\right). \quad (19)$$

In the case of orthonormal signals, $(\{\beta_{i,j}\}_{i,j=1, i \neq j}^M = 0)$ and the eigenvalues of the matrix $\mathbf{R}_s = \mathbf{I}_M$ are equal to one. Note that in this case, the matrix of eigenvectors $\mathbf{U}$ is not unique. A simple choice for $\mathbf{U}$ is $\mathbf{U} = \mathbf{I}_M$. Therefore

$$\mathbf{d}_{\beta=0}(\theta) = \sqrt{N}\text{vec}(\mathbf{A}(\theta)\mathbf{I}_M) = \sqrt{N}\text{vec}(\mathbf{a}(\theta)\mathbf{a}^T(\theta)). \quad (20)$$

The equivalent steering vector for orthonormal signals is the product of the steering vector in the receive mode $\mathbf{a}(\theta)$ and the steering vector in the transmit mode $\mathbf{a}^T(\theta)$. The equivalent steering vector for noncoherent signals includes all the M^2 elements of $\mathbf{a}(\theta)\mathbf{a}^T(\theta)$, which represent all the possible transmit–receive combinations. The im th element of this matrix is $a_i(\theta)a_m(\theta) = \exp(-jk^T(\theta)(\mathbf{x}_i + \mathbf{x}_m))$. Hence, the array response consists of virtual sensors located at the combinations of $\mathbf{x}_i + \mathbf{x}_m$ for $i, m = 1, \dots, M$. Consequently, the array aperture is virtually extended. This virtual aperture extension

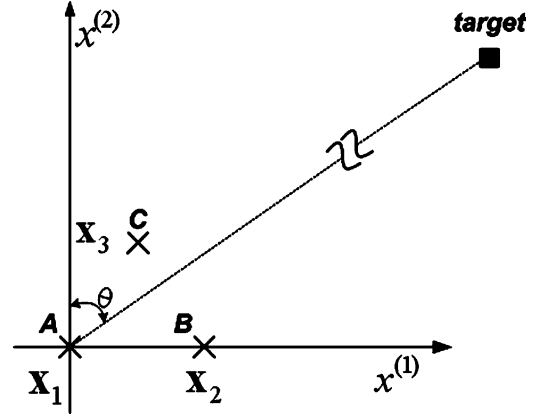


Fig. 3. Array aperture for coherent signals $M = 3, L = 1$.

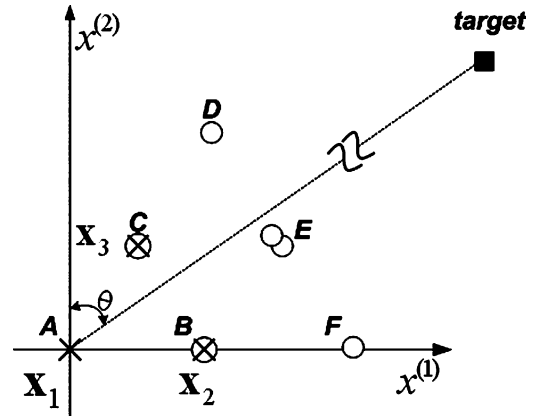


Fig. 4. Array aperture for orthogonal signals $M = 3, L = 1$; $\times$ -points: actual sensors, $\circ$ -points: virtual sensors. Two sensors are located at points B, C, E.

results in narrower beams and therefore higher angular resolution and better detection performance. Moreover, some of the virtual sensor locations are identical, which can be interpreted as spatial tapering and results in lower sidelobes.

In order to illustrate these advantages, we examine an example with three array elements ($M = 3$), which are located at vertices of an equilateral triangle (see $\times$ -points in Fig. 3) and one target ($L = 1$). In Figs. 3 and 4, the equivalent array structure for coherent and orthogonal signals is presented, respectively. As mentioned above, the equivalent array for orthogonal signals includes all the transmit–receive combinations of the elements of $\mathbf{a}(\theta)\mathbf{a}^T(\theta)$, which are given by $\{\exp(-jk^T(\theta)(\mathbf{x}_i + \mathbf{x}_m))\}_{i,m=1}^M$. This is equivalent to an extended array, whose elements are located at $\{\mathbf{x}_i + \mathbf{x}_m\}_{i,m=1}^M$. Therefore the equivalent array consists of virtual sensors (marked by $\circ$ -points in Fig. 4) in addition to the actual sensors.

According to (3), the total number of delays is nine, representing nine different virtual sensors. In this configuration, we obtain three sensors at the actual sensor locations (points A, B, C), three virtual sensors at new locations (points D, E, F), and three additional virtual sensors, which fall at the locations of other sensors (points B, C, E). The last three sensors, as discussed above, can be interpreted as spatial weighting or tapering. Finally, the virtual aperture of the orthogonal signals is

created by nine sensors (six of them are at different locations), compared to three sensors of the coherent signals model.

B. Spatial Coverage Extension

In conventional target detection and localization systems, several directional beams are usually transmitted in order to scan a given region of interest (ROI). Each directional beam is generated using coherent transmitted signals. The time-on-target (TOT) for each transmitted beam is equal to the total interval assigned for covering the ROI, divided by the number of beams required to cover the given ROI. When the transmitted signals are orthogonal, the beams become omnidirectional, causing reduction in the beam gain. On the other hand, transmission of omnidirectional beams extends the spatial coverage of each beam; therefore, it allows increase of the TOT interval for each beam. In fact, the TOT interval for orthogonal signals is equal to the interval which is assigned to scan the ROI. Therefore, the beam gain loss can be compensated by increased TOT. Instead of M directional beams, one omnidirectional beam can be transmitted with M times higher TOT interval. Hence, in the sequel, when comparing the spatially orthogonal and coherent signals, this TOT compensation will be considered.

Furthermore, the spatial transmitted power density with orthogonal signals is constant at each direction, while in the coherent signal case, the spatial transmitted power density is nonuniform and depends on the beam shape and the overlap between the beams.

In omnidirectional signal transmission, the echo signal should be processed for a larger ROI, and therefore statistically a larger number of targets. If the targets in the given ROI are disjoint in range or Doppler, then for each range-Doppler bin, a single target case should be considered. In practice, the probability of multiple targets in a given range-Doppler bin is low, although it is still higher than the case of directional signal transmission in which only the targets within the narrow beam are excited.

C. Beam Pattern Improvement

The transmit-receive pattern can be written as

$$G_{\text{TR}}(\theta) = \frac{|\mathbf{d}^H(\theta)\mathbf{d}(\theta_o)|^2}{\|\mathbf{d}(\theta)\|^2} \quad (21)$$

where θ_o is the target DOA and θ is the digital beam direction. Equation (21) can be rewritten in terms of the steering vector $\mathbf{a}(\theta)$ (see Appendix B)

$$G_{\text{TR}}(\theta) = \frac{N |\mathbf{a}^H(\theta)\mathbf{R}_s^T \mathbf{a}(\theta_o)|^2 |\mathbf{a}^H(\theta)\mathbf{a}(\theta_o)|^2}{M \mathbf{a}^H(\theta)\mathbf{R}_s^T \mathbf{a}(\theta)} \quad (22)$$

It is worthwhile to notice that the right term in the numerator of (22) represents the beam pattern in the receive mode and is independent of the transmitted signal coherence matrix. The other terms in (22) represent the beam pattern in the transmit mode. For coherent transmit signals steered to the array broadside with ($\mathbf{R}_s = \mathbf{1}\mathbf{1}^T$), the transmit-receive pattern is given by

$$G_{\text{TR}}^{\beta=1}(\theta) = \frac{N |\mathbf{a}^H(\theta_o)\mathbf{1}|^2 |\mathbf{a}^H(\theta)\mathbf{a}(\theta_o)|^2}{M} \quad (23)$$

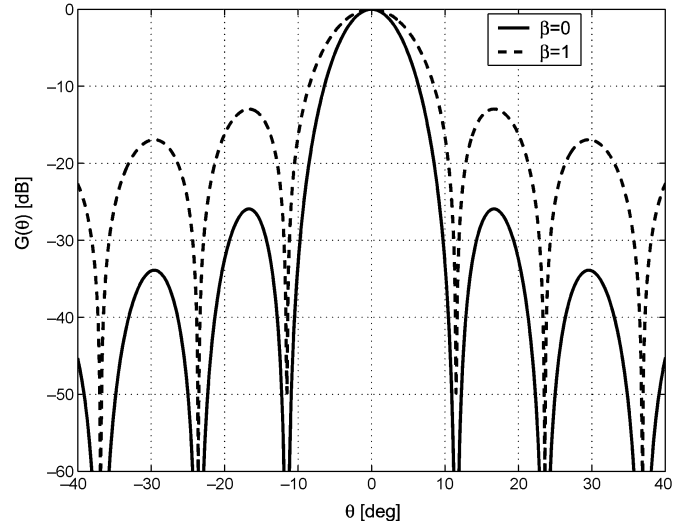


Fig. 5. Beam pattern for orthogonal and coherent signals with a ULA of $M = 10$ elements and with half a wavelength spacing.

and for orthonormal signals ($\mathbf{R}_s = \mathbf{I}_M$) considering the increased TOT by factor M

$$G_{\text{TR}}^{\beta=0}(\theta) = \frac{N |\mathbf{a}^H(\theta)\mathbf{a}(\theta_o)|^4}{M} \quad (24)$$

The term $|\mathbf{a}^H(\theta_o)\mathbf{1}|^2$ in (23) introduces the attenuation in the transmit gain due to beam shape loss. This attenuation does not exist for the orthogonal signals case. The beam patterns for coherent and orthogonal transmitted signals are shown in Fig. 5. The array is uniform linear array (ULA) with $M = 10$ elements and with half a wavelength spacing, where the transmit beam of coherent signals is directed to $\theta_o = 0^\circ$, i.e., the phases of $\{\beta_{im} = 0\}_{i,m=1}^M$. Fig. 5 shows that the orthogonal transmitted signal model provides narrower beam width by $\sqrt{2}$ and lower sidelobe levels, compared to the coherent signal model. Note that the sidelobe level in the coherent signal case is at about -13 dB below the mainlobe level, which reflects the contribution of the digital beamforming in the receive mode only. In the orthogonal signal case, the sidelobe level is at about -26 dB below the mainlobe level reflecting the contribution of the digital beamforming in the transmission in addition to the receive mode. These phenomena can also be interpreted as the contribution of the virtual aperture extension and virtual tapering, as mentioned above. In Fig. 6, the beam patterns for coherent and orthogonal signals are shown, where the target is located at $\theta_o = 0^\circ, 5^\circ, 10^\circ$, respectively. It can be observed that for coherent transmitted signals, the gain of the transmit-receive pattern is attenuated when the target is not located in the center of the transmit beam, because of the beam shape loss. However, for orthogonal transmitted signals, the gain remains constant for all target directions θ_o .

D. Increase of the Limit on the Number of Targets

In conventional localization methods, an array of M elements allows detection of up to $L = M - 1$ targets with unknown complex amplitudes. In the spatially coded signal model, as discussed above, the number of different virtual sensors is the

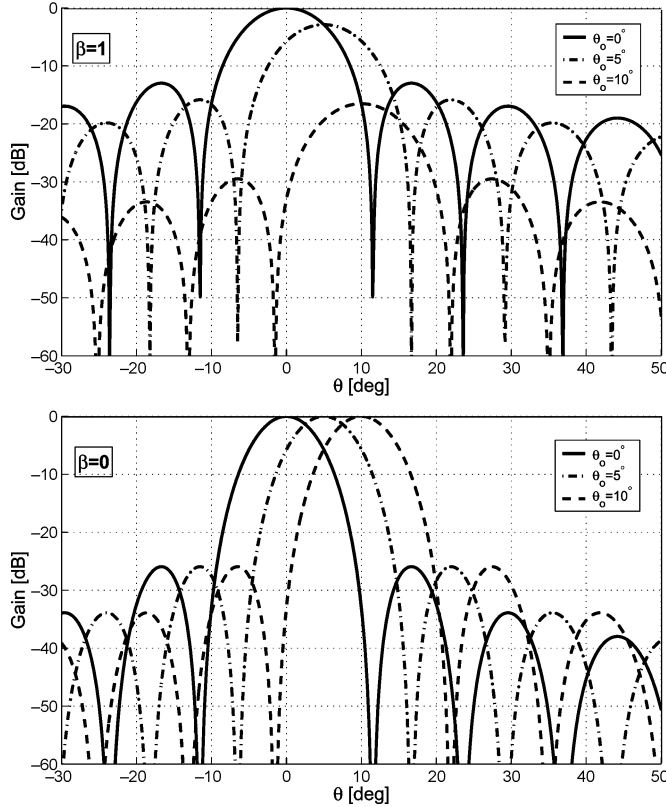


Fig. 6. Beam pattern for orthogonal and coherent signals with a ULA of $M = 10$ elements and with half a wavelength spacing. The beam in the transmit mode is directed to $\theta = 0^\circ$ and the target is located at $\theta_o = 0^\circ, 5^\circ, 10^\circ$.

number of different combinations of transmit–receive delays $\tau_{im}(\theta)$ according to (3). Hence, the limit on the maximum number of targets is now constrained by $L \leq M(M+1)/2 - 1$. This upper bound on the number of targets can be smaller depending on the array geometry. Thus, orthogonal signal transmission significantly increases the upper limit on the number of targets that can be localized.

V. TARGET DETECTION AND DOA ESTIMATION

In order to demonstrate the advantages of the proposed configuration in target detection and localization, the ML and CRB on DOA estimation and GLRT for target detection are derived.

A. Maximum Likelihood Estimation

The model in (15) can be rewritten in matrix form

$$\tilde{\mathbf{y}} = \mathbf{D}(\boldsymbol{\theta})\boldsymbol{\alpha} + \mathbf{v} \quad (25)$$

where $\mathbf{D}(\boldsymbol{\theta}) = [\mathbf{d}(\theta_1), \dots, \mathbf{d}(\theta_L)]$ and $\boldsymbol{\alpha} = [\alpha_1, \dots, \alpha_L]^T$. Hence, the ML estimator for target localization for the model in (25) is given by

$$(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\alpha}})_{\text{ML}} = \arg \min_{\boldsymbol{\theta}, \boldsymbol{\alpha}} \|\tilde{\mathbf{y}} - \mathbf{D}(\boldsymbol{\theta})\boldsymbol{\alpha}\|^2. \quad (26)$$

After optimization with respect to $\boldsymbol{\alpha}$, the ML estimator for $\boldsymbol{\theta}$ is given by

$$\hat{\boldsymbol{\theta}}_{\text{ML}} = \arg \max_{\boldsymbol{\theta}} L(\boldsymbol{\theta}) \quad (27)$$

$$L(\boldsymbol{\theta}) \triangleq \tilde{\mathbf{y}}^H \mathbf{P}_{\mathbf{D}}(\boldsymbol{\theta}) \tilde{\mathbf{y}}, \quad (28)$$

where $\mathbf{P}_{\mathbf{D}}(\boldsymbol{\theta}) \triangleq \mathbf{D}(\boldsymbol{\theta})(\mathbf{D}^H(\boldsymbol{\theta})\mathbf{D}(\boldsymbol{\theta}))^{-1}\mathbf{D}^H(\boldsymbol{\theta})$ is a projection matrix into the subspace spanned by columns of $\mathbf{D}(\boldsymbol{\theta})$. For the scenario of a single target, $L(\boldsymbol{\theta})$ can be rewritten as

$$L(\boldsymbol{\theta}) = \tilde{\mathbf{y}}^H \mathbf{P}_{\mathbf{d}}(\boldsymbol{\theta}) \tilde{\mathbf{y}} = \frac{|\mathbf{d}_{\boldsymbol{\beta}}^H(\boldsymbol{\theta}) \tilde{\mathbf{y}}|^2}{\|\mathbf{d}_{\boldsymbol{\beta}}(\boldsymbol{\theta})\|^2} \quad (29)$$

where $\mathbf{P}_{\mathbf{d}}(\boldsymbol{\theta})$ is a projection matrix into the subspace spanned by columns of $\mathbf{d}_{\boldsymbol{\beta}}(\boldsymbol{\theta})$. By substitution of $\tilde{\mathbf{y}}$ from (12), $\mathbf{d}_{\boldsymbol{\beta}}(\boldsymbol{\theta})$ from (16), and using the identity $\text{tr}(XY) = (\text{vec}(X^H))^H \text{vec}(Y)$ (see [12]), with some trace properties, the numerator of (29) can be expressed as

$$|\mathbf{d}_{\boldsymbol{\beta}}^H(\boldsymbol{\theta}) \tilde{\mathbf{y}}|^2 = N |\text{tr}(\mathbf{A}^H(\boldsymbol{\theta})\mathbf{E})|^2 = N |\mathbf{a}^H(\boldsymbol{\theta})\mathbf{E}\mathbf{a}^*(\boldsymbol{\theta})|^2 \quad (30)$$

where $\mathbf{E}$ is the sufficient statistic matrix, defined in (10). In Appendix B, it is shown that

$$\|\mathbf{d}_{\boldsymbol{\beta}}(\boldsymbol{\theta})\|^2 = N M \mathbf{a}^H(\boldsymbol{\theta}) \mathbf{R}_s^T \mathbf{a}(\boldsymbol{\theta}). \quad (31)$$

Hence, (27) for a single target becomes

$$L(\boldsymbol{\theta}) = \frac{|\mathbf{a}^H(\boldsymbol{\theta})\mathbf{E}\mathbf{a}^*(\boldsymbol{\theta})|^2}{M \mathbf{a}^H(\boldsymbol{\theta}) \mathbf{R}_s^T \mathbf{a}(\boldsymbol{\theta})} \quad (32)$$

and the ML estimator of $\boldsymbol{\theta}$ can be written as

$$\hat{\boldsymbol{\theta}}_{\text{ML}} = \arg \max_{\boldsymbol{\theta}} \frac{|\mathbf{a}^H(\boldsymbol{\theta})\mathbf{E}\mathbf{a}^*(\boldsymbol{\theta})|^2}{M \mathbf{a}^H(\boldsymbol{\theta}) \mathbf{R}_s^T \mathbf{a}(\boldsymbol{\theta})}. \quad (33)$$

B. Detection

The hypotheses for a single target detection for the sufficient statistic model in (15), can be stated as

$$\begin{aligned} H_0 : \tilde{\mathbf{y}} &= \mathbf{v} \\ H_1 : \tilde{\mathbf{y}} &= \alpha \mathbf{d}_{\boldsymbol{\beta}}(\boldsymbol{\theta}) + \mathbf{v}. \end{aligned} \quad (34)$$

Thus, the GLRT is given by

$$L_{\tilde{\mathbf{y}}} \triangleq \max_{\boldsymbol{\theta}, \alpha} \{\log f_{\tilde{\mathbf{y}}}(\tilde{\mathbf{y}}, \boldsymbol{\theta}, \alpha; H_1)\} - \log f_{\tilde{\mathbf{y}}}(\tilde{\mathbf{y}}; H_0) \underset{H_0}{\overset{H_1}{\gtrless}} \delta \quad (35)$$

where $f_{\tilde{\mathbf{y}}}(\tilde{\mathbf{y}}, \boldsymbol{\theta}, \alpha; H_1)$ and $f_{\tilde{\mathbf{y}}}(\tilde{\mathbf{y}}; H_0)$ are the probability density functions of the sufficient statistic under hypotheses H_1 and H_0 , respectively. Hence, the GLRT for the model in (6) can be written as

$$L(\hat{\boldsymbol{\theta}}_{\text{ML}}) = \frac{|\mathbf{a}^H(\hat{\boldsymbol{\theta}}_{\text{ML}})\mathbf{E}\mathbf{a}^*(\hat{\boldsymbol{\theta}}_{\text{ML}})|^2}{M \mathbf{a}^H(\hat{\boldsymbol{\theta}}_{\text{ML}}) \mathbf{R}_s^T \mathbf{a}(\hat{\boldsymbol{\theta}}_{\text{ML}})} \underset{H_0}{\overset{H_1}{\gtrless}} \delta. \quad (36)$$

The threshold δ is set according to the desired false alarm rate. It is interesting to find the asymptotic statistics of $L(\hat{\boldsymbol{\theta}}_{\text{ML}})$ for

both threshold setting and for performance evaluation. Unfortunately, under hypothesis H_0 , the target parameters θ cannot be estimated, and thus derivation of the asymptotic properties of the above GLRT is a difficult problem [13]. However, it can be verified that for orthogonal signals, there is no coupling between α and θ . Therefore, the fact that θ is unknown does not affect the estimation performance of α , which is required for decision between the two hypotheses. Accordingly, for the orthogonal signals case, the asymptotic statistics of $L(\hat{\theta}_{ML})$ under the two hypotheses are given by [10]

$$L(\hat{\theta}_{ML}) \sim \begin{cases} \chi_2^2, & H_0 \\ \chi_2'^2(\rho), & H_1 \end{cases} \quad (37)$$

where χ_2^2 and $\chi_2'^2(\rho)$ are central and noncentral chi-squared distributions with two degrees of freedom, respectively, and ρ is the noncentrality parameter, which is equal to

$$\rho = \frac{\|\alpha \mathbf{d}_\beta(\theta)\|^2}{\sigma_w^2} = \frac{|\alpha|^2}{\sigma_w^2} |\mathbf{a}^H(\theta) \mathbf{R}_s^T \mathbf{a}(\theta)|^2 = \frac{|\alpha|^2 M^2}{\sigma_w^2}. \quad (38)$$

Using the Neyman–Pearson criterion, the probability of false alarm is $P_{FA} = Q_{\chi_2^2}(\delta/\sigma_w) = \text{const}$ and the probability of detection is $P_D = Q_{\chi_2'^2(\rho)}(\delta/\sigma_w)$, where $Q_{\chi_2^2}(\cdot)$ and $Q_{\chi_2'^2(\rho)}(\cdot)$ are right-tail probability functions [10].

C. Cramér–Rao Bound

The Fisher information matrix (FIM) [9] for estimating the vector $\boldsymbol{\xi} = [\theta^T \ \tilde{\alpha}^T]^T$ from the sufficient statistic $\tilde{\mathbf{y}}$, according to the model in (7), can be partitioned as

$$\mathbf{J}(\tilde{\mathbf{y}}) = \begin{bmatrix} \mathbf{J}_{\theta\theta} & \mathbf{J}_{\theta\tilde{\alpha}} \\ \mathbf{J}_{\tilde{\alpha}\theta}^T & \mathbf{J}_{\tilde{\alpha}\tilde{\alpha}} \end{bmatrix} \quad (39)$$

and the CRB for the DOA estimate can be expressed as

$$\text{CRB}(\theta) = [\mathbf{J}_{\theta\theta} - \mathbf{J}_{\theta\tilde{\alpha}} \mathbf{J}_{\tilde{\alpha}\tilde{\alpha}}^{-1} \mathbf{J}_{\tilde{\alpha}\theta}^T]^{-1} \quad (40)$$

where $\mathbf{J}_{\theta\theta}$, $\mathbf{J}_{\theta\tilde{\alpha}}$, and $\mathbf{J}_{\tilde{\alpha}\tilde{\alpha}}$ are matrices, whose elements are derived in Appendix C

$$J_{\theta_l \theta_p} = \frac{2N}{\sigma_w^2} \text{Re} \left\{ \alpha_l^* \alpha_p \text{tr} \left(\dot{\mathbf{A}}(\theta_p) \mathbf{R}_s \dot{\mathbf{A}}^H(\theta_l) \right) \right\} \quad (41)$$

$$\mathbf{J}_{\tilde{\alpha}_l \tilde{\alpha}_p} = \frac{2N}{\sigma_w^2} \text{Re} \{ (1, j)^H (1, j) \text{tr} (\mathbf{A}(\theta_p) \mathbf{R}_s \mathbf{A}^H(\theta_l)) \} \quad (42)$$

$$\mathbf{J}_{\theta_l \tilde{\alpha}_p} = \frac{2N}{\sigma_w^2} \text{Re} \left\{ \alpha_l^* \text{tr} \left(\mathbf{A}(\theta_p) \mathbf{R}_s \dot{\mathbf{A}}^H(\theta_l) \right) (1, j) \right\}. \quad (43)$$

In Appendix C, it is shown that if the array origin is chosen to be at the array centroid, i.e., $\sum_{m=1}^M \mathbf{x}_m = 0$, the CRB for DOA estimation of a single target is given by (44) as shown at the bottom of the page, where $\text{SNR} \triangleq N|\alpha|^2/\sigma_w^2$.

For the case of two elements ($M = 2$) with $\angle\beta = 0$ (i.e., the transmit beam is steered to the array broadside), the CRB for

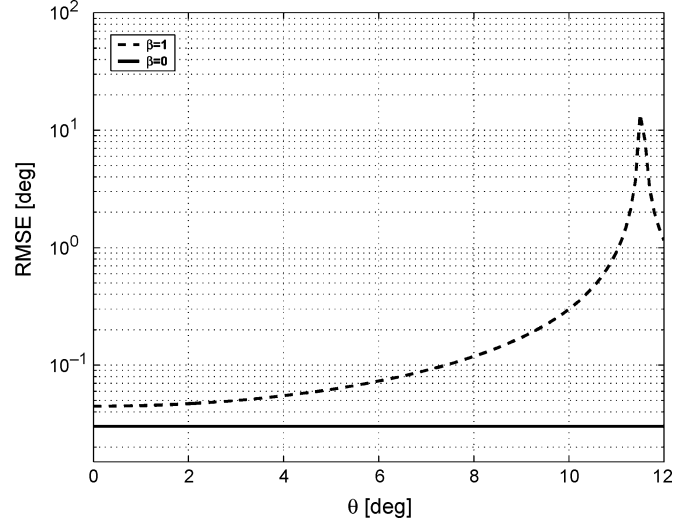


Fig. 7. CRB on DOA for $M = 10$, $L = 1$, $\text{SNR} = 0$ dB. The target is located at $\theta_t = 0^\circ$ and the beam is steered at θ for coherent signals.

DOA estimation can be expressed as a function of $\beta = |\beta_{12}|$ by inserting $\mathbf{a}(\theta) = [1 \ \exp(j2\pi(d/\lambda) \sin \theta)]^T$

$$\text{CRB}(\theta)_\beta = \frac{1}{2\text{SNR}} \frac{1}{e_c^2 \left(2 - \frac{\beta^2 \sin^2 e_s}{1 + \beta \cos e_s} \right)} \quad (45)$$

where $e_s \triangleq 2\pi(d/\lambda) \sin \theta$, $e_c \triangleq 2\pi(d/\lambda) \cos \theta$, and d is the distance between the elements. The optimal value of β , which minimizes the CRB, can be obtained by differentiating (45) with respect to β and then equating to zero. The minimal CRB for $M = 2$ is obtained for $\beta = 0$ even without TOT compensation, which represents the case of orthonormal transmitted signals. Furthermore, it can be shown that

$$\text{CRB}(\theta)_{\beta=0} \leq \text{CRB}(\theta)_\beta \quad \forall \theta \quad (46)$$

where equality is satisfied only for $\theta = 0^\circ$. From (46), we conclude that the DOA estimation performance with orthogonal transmitted signals is superior to the performance obtained with coherent transmitted signals. Note that for $\beta = 0$, the TOT can be compensated by a factor of $M = 2$, and thus the bound can further be decreased.

VI. SIMULATION RESULTS

In this section, we demonstrate via simulations the detection and localization performance for the case of spatially coded signals. A ULA with half a wavelength spacing is considered.

In Fig. 7, the CRB for DOA estimation root-mean-square error (RMSE) of a single target ($L = 1$) as a function of its direction θ is plotted for coherent ($\beta = 1$) and orthonormal

$$\text{CRB}(\theta) = \frac{1}{2\text{SNR} \left(M \dot{\mathbf{a}}^H(\theta) \mathbf{R}_s^T \dot{\mathbf{a}}(\theta) + \mathbf{a}^H(\theta) \mathbf{R}_s^T \mathbf{a}(\theta) \|\dot{\mathbf{a}}(\theta)\|^2 - \frac{M |\mathbf{a}^H(\theta) \mathbf{R}_s^T \dot{\mathbf{a}}(\theta)|^2}{\mathbf{a}^H(\theta) \mathbf{R}_s^T \mathbf{a}(\theta)} \right)} \quad (44)$$

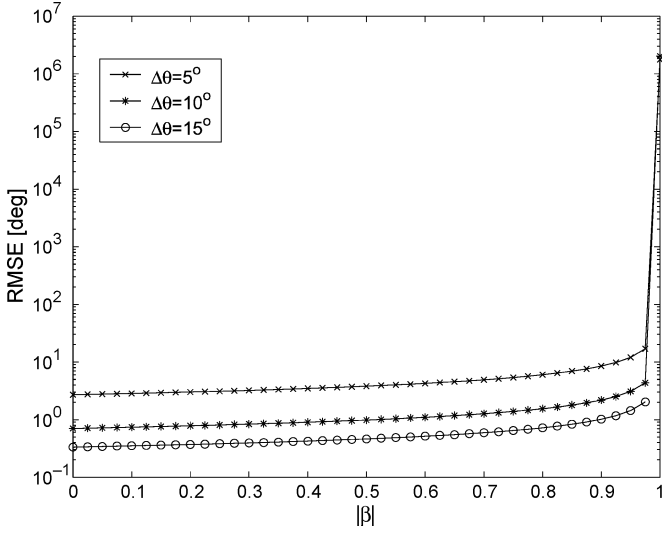


Fig. 8. CRB on DOA estimation, $L = 2$, $M = 2$, $\text{SNR} = 0$ dB; $\theta_1 = 0^\circ$, $\theta_2 = 5^\circ, 10^\circ, 15^\circ$. Two targets can be resolved by two elements if incoherent signals are transmitted.

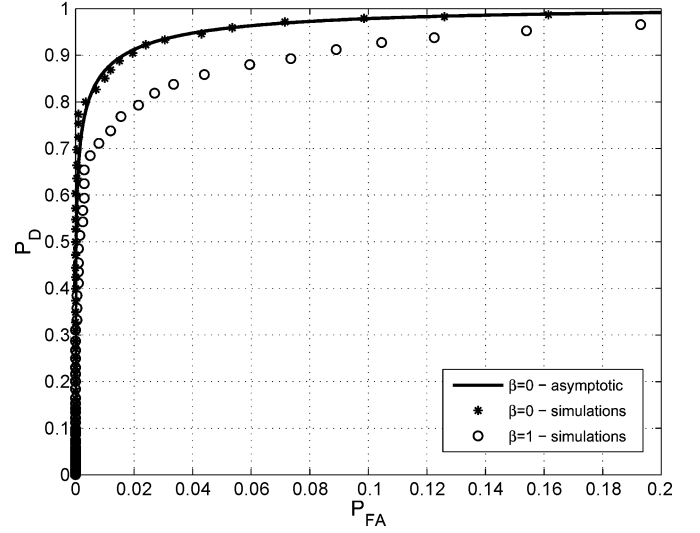


Fig. 10. ROC for the coherent and orthogonal transmitted signals; $M = 10$, $L = 1$, $\theta = 0^\circ$, and $\text{SNR} = 10$ dB. For the coherent transmitted signals, the beam is directed at $\theta = 0^\circ$.

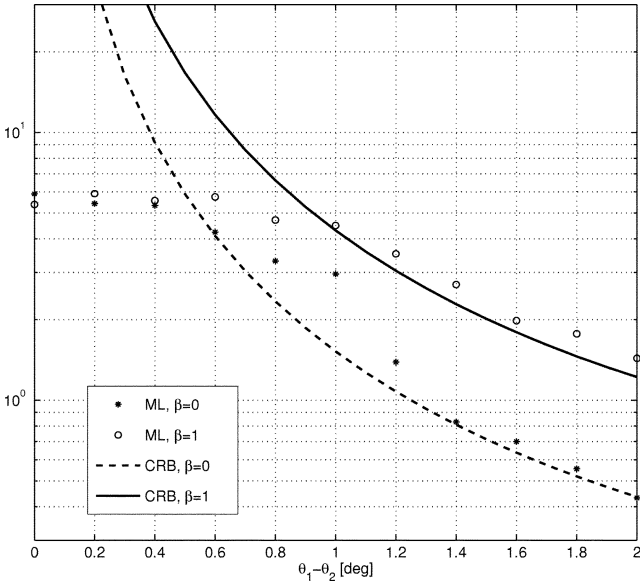


Fig. 9. CRB and ML performance for DOA estimation of the first target for coherent and orthogonal transmitted signals; $M = 10$ elements, $L = 2$ targets; $\theta_1 = 0^\circ$, $\theta_2 \in [0, 2]^\circ$, $\text{SNR} = 0$ dB.

signals ($\beta = 0$). The array includes $M = 10$ elements and $\text{SNR} = 0$ dB. Note that for orthogonal transmitted signals, the CRB is constant with respect to θ , while in the coherent transmitted signals case, the CRB increases with θ due to the beam shape loss.

Fig. 8 presents the CRB for localization of a target located at $\theta_1 = 0^\circ$ in the presence of an additional target at $\theta_2 = 5^\circ, 10^\circ, 15^\circ$ with an array of $M = 2$ elements. Obviously, the condition $L \leq M - 1$ which is relevant to the case of coherent signals $|\beta| = 1$ is not satisfied. However, Fig. 8 shows that the CRB is finite for $|\beta| < 1$. This phenomenon is directly related to the contribution of the virtual sensors in the noncoherent signals case, as mentioned in Section IV. In this case, the number of

virtual sensors at the different locations is $M(M + 1)/2 = 3$, and thus localization of two targets is possible with an array of two elements. As expected, the CRB goes to infinity for $|\beta| \rightarrow 1$.

In Fig. 9, the angular resolution with coherent and orthogonal transmitted signals is examined. The scenario includes $M = 10$ elements and $L = 2$ targets, where the first target is located at $\theta_1 = 0^\circ$. The CRB and performance of the ML estimator of the first target angle as a function of the position of the second target θ_2 for both coherent and orthogonal transmitted signals are depicted. The TOT compensation is taken into consideration for the orthogonal transmitted signals. It can be noticed that the performance of the spatially coded signal model with orthogonal transmitted signals is superior to the configuration with coherent transmitted signals.²

In Fig. 10, receiver operating characteristic (ROC) curves are presented using simulation results for both coherent and orthogonal transmitted signals. The theoretical asymptotic detection performance for the case of orthogonal signals is also calculated using (37) and depicted in this figure. The scenario includes $M = 10$ elements and $L = 1$ target, which is located at $\theta = 20^\circ$, and for the coherent transmitted signals the beam is directed at $\theta = 0^\circ$. The TOT compensation is taken into consideration for the orthogonal transmitted signals. It can be observed that the detection performance with orthogonal signals is higher than with coherent signals. In addition, the asymptotic performance obtained theoretically coincides with the simulations for the case of orthogonal signals.

VII. DISCUSSION AND CONCLUSIONS

In this paper, a new approach for space-time signal transmission in radar and sonar systems was presented. Spatially orthogonal signal transmission enables digital beamforming and array

²At low angular separation, the ML estimation error RMSE is lower than the CRB because of the limited search region for estimation of θ_1 and θ_2 .

processing of the transmitted signal in addition to the conventional array processing in the receive mode. The ability to digitally steer the transmitted beam, allows one to avoid possible beam shape loss when the target is not located in the center of the transmitted beam. The proposed method allows one to obtain M^2 virtual sensors instead of M real sensors. These virtual sensors extend the array aperture, which results in narrower beamwidth. The fact that part of the virtual sensors fall in the same locations can be interpreted as virtual tapering of the array, which results in lower sidelobes. These phenomena are directly related to higher performance in target detection, localization accuracy, and angular resolution. Another benefit from the virtual sensors is increasing the upper limit on the number of targets that can be detected and localized, that is, $M(M+1)/2-1$ targets in the cases of incoherent transmitted signals versus $M-1$ in the case of coherent transmitted signals. The detection and localization performances for orthogonal and coherent transmitted signals were evaluated using simulations and compared to the theory. The results showed superior performance of the orthogonal signals compared to coherent signals.

Spatially orthogonal signal transmission reduces the spatial transmitted power density by M . This is of great importance in many military applications, which require low detectability of the active system. Moreover, lower spatial transmitted power density reduces the interference to neighboring systems.

In arrays with large number of elements, multiple matched filtering for each element, as presented in Fig. 2, may be complicated, expensive, and impractical. In such systems the new configuration may be implemented by division of array of elements into subarrays, transmitting different signals. This is equivalent to the case where the signal coherence matrix in (1) is block diagonal, in which each block corresponds to a different subarray. Consequently, the number of matched filters can be reduced at the cost of performance reduction.

Implementation of methods for multiple target localization, like MUSIC, on the model described in (7) is straightforward. Another approach for handling multiple targets or interference, which is simple to implement, is adaptive beamforming, such as Capon's beamformer. Possible clutter and jammer can be treated as interference sources which can be suppressed using these methods.

APPENDIX A LOG-LIKELIHOOD DERIVATION

Consider the data sequence $\mathbf{Y} = (\mathbf{y}[1], \dots, \mathbf{y}[N])$, whose elements are statistically independent with complex Gaussian distribution $\mathbf{y}[n] \sim N^c(\sum_{l=1}^L \alpha_l \mathbf{A}(\theta_l) \mathbf{s}[n], \sigma_w^2 \mathbf{I}_M)$. The log-likelihood function for estimating $\boldsymbol{\xi}$ from $\mathbf{Y}$ is given by

$$\log f_{\mathbf{Y}}(\mathbf{Y}; \boldsymbol{\xi}) = h(\mathbf{Y}) + g_1(\mathbf{Y}, \boldsymbol{\xi}) + g_2(\boldsymbol{\xi}) \quad (47)$$

where

$$h(\mathbf{Y}) = -MN \log(\pi \sigma_w^2) - \frac{1}{\sigma_w^2} \sum_{n=1}^N \|\mathbf{y}[n]\|^2 \quad (48)$$

$$g_1(\mathbf{Y}, \boldsymbol{\xi}) = \frac{2}{\sigma_w^2} \text{Re} \left\{ \sum_{l=1}^L \alpha_l^* \sum_{n=1}^N \mathbf{s}^H[n] \mathbf{A}^H(\theta_l) \mathbf{y}[n] \right\} \quad (49)$$

$$g_2(\boldsymbol{\xi}) = -\frac{1}{\sigma_w^2} \sum_{n=1}^N \left\| \sum_{l=1}^L \alpha_l \mathbf{A}(\theta_l) \mathbf{s}[n] \right\|^2. \quad (50)$$

According to the Neyman–Fisher factorization [9], the function $g_1(\cdot, \cdot)$ depends on the data $\mathbf{Y}$ only through the sufficient statistic. Denoting the columns of $\mathbf{A}(\theta)$ by $\{\mathbf{A}_{:m}(\theta)\}_{m=1}^M$, we obtain

$$g_1(\mathbf{Y}, \boldsymbol{\xi}) = \frac{2\sqrt{N}}{\sigma_w^2} \text{Re} \left\{ \sum_{l=1}^L \alpha_l^* \sum_{m=1}^M \mathbf{A}_{:m}^H(\theta_l) \times \left(\frac{1}{\sqrt{N}} \sum_{n=1}^N \mathbf{y}[n] \mathbf{s}_m^*[n] \right) \right\}. \quad (51)$$

The term $g_2(\boldsymbol{\xi})$ becomes

$$g_2(\boldsymbol{\xi}) = -\frac{1}{\sigma_w^2} \sum_{n=1}^N \sum_{k,l=1}^L \alpha_k^* \alpha_l \mathbf{s}^H[n] \mathbf{A}^H(\theta_k) \mathbf{A}(\theta_l) \mathbf{s}[n] \\ = -\frac{N}{\sigma_w^2} \sum_{k,l=1}^L \alpha_k^* \alpha_l \text{tr}(\mathbf{A}(\theta_l) \mathbf{R}_s \mathbf{A}^H(\theta_k)). \quad (52)$$

By substitution of (5), (52) can be rewritten as

$$g_2(\boldsymbol{\xi}) = -\frac{N}{\sigma_w^2} \sum_{k,l=1}^L \alpha_k^* \alpha_l \text{tr}(\mathbf{a}(\theta_l) \mathbf{a}^T(\theta_l) \mathbf{R}_s \mathbf{a}^*(\theta_k) \mathbf{a}^H(\theta_k)) \\ = -\frac{N}{\sigma_w^2} \sum_{k,l=1}^L \alpha_k^* \alpha_l \mathbf{a}^T(\theta_l) \mathbf{R}_s \mathbf{a}^*(\theta_k) \mathbf{a}^H(\theta_k) \mathbf{a}(\theta_l). \quad (53)$$

Finally, the log-likelihood function in (8) is obtained by substitution of (48), (51), and (53) into (47).

APPENDIX B TRANSMIT–RECEIVE PATTERN DERIVATION

By inserting (16) into (21), we obtain

$$G_{\text{TR}}(\theta) = \frac{\left| \sqrt{N} \text{vec}(\mathbf{A}(\theta) \mathbf{U} \mathbf{\Lambda}^{1/2})^H \sqrt{N} \text{vec}(\mathbf{A}(\theta_o) \mathbf{U} \mathbf{\Lambda}^{1/2}) \right|^2}{\left\| \sqrt{N} \text{vec}(\mathbf{A}(\theta) \mathbf{U} \mathbf{\Lambda}^{1/2}) \right\|^2}. \quad (54)$$

Using the identity [12] $\text{tr}(XY) = (\text{vec}(X^H))^H \text{vec}(Y)$, (54) can be rewritten as

$$G_{\text{TR}}(\theta) = \frac{N \left| \text{tr}(\mathbf{\Lambda}^{1/2} \mathbf{U}^H \mathbf{A}^H(\theta) \mathbf{A}(\theta_o) \mathbf{U} \mathbf{\Lambda}^{1/2}) \right|^2}{\text{tr}(\mathbf{\Lambda}^{1/2} \mathbf{U}^H \mathbf{A}^H(\theta) \mathbf{A}(\theta) \mathbf{U} \mathbf{\Lambda}^{1/2})} \\ = \frac{N \left| \text{tr}(\mathbf{A}(\theta_o) \mathbf{R}_s \mathbf{A}^H(\theta)) \right|^2}{\text{tr}(\mathbf{A}(\theta) \mathbf{R}_s \mathbf{A}^H(\theta))}. \quad (55)$$

By substitution of (5) into (55), the transmit–receive pattern can be expressed as

$$G_{\text{TR}}(\theta) = \frac{N \left| \mathbf{a}^T(\theta_o) \mathbf{R}_s \mathbf{a}^*(\theta) \mathbf{a}^H(\theta) \mathbf{a}(\theta_o) \right|^2}{\mathbf{a}^T(\theta) \mathbf{R}_s \mathbf{a}^*(\theta) \mathbf{a}^H(\theta) \mathbf{a}(\theta)} \\ = \frac{N \left| \mathbf{a}^H(\theta) \mathbf{R}_s^T \mathbf{a}(\theta_o) \right|^2 \left| \mathbf{a}^H(\theta) \mathbf{a}(\theta_o) \right|^2}{M \mathbf{a}^H(\theta) \mathbf{R}_s^T \mathbf{a}(\theta)}. \quad (56)$$

$$\begin{aligned} \text{CRB}(\theta) &= [J_{\theta\theta} - \mathbf{J}_{\theta\tilde{\alpha}} \mathbf{J}_{\tilde{\alpha}\tilde{\alpha}}^{-1} \mathbf{J}_{\tilde{\alpha}\theta}^T]^{-1} \\ &= \frac{\text{tr}(\mathbf{A}(\theta) \mathbf{R}_s \mathbf{A}^H(\theta))}{2\text{SNR} \left(\text{tr}(\dot{\mathbf{A}}(\theta) \mathbf{R}_s \dot{\mathbf{A}}^H(\theta)) \text{tr}(\mathbf{A}(\theta) \mathbf{R}_s \mathbf{A}^H(\theta)) - \left| \text{tr}(\mathbf{A}(\theta) \mathbf{R}_s \dot{\mathbf{A}}^H(\theta)) \right|^2 \right)} \end{aligned} \quad (63)$$

$$\text{CRB}(\theta) = \frac{1}{2\text{SNR} \left(M \dot{\mathbf{a}}^H(\theta) \mathbf{R}_s^T \dot{\mathbf{a}}(\theta) + \mathbf{a}^H(\theta) \mathbf{R}_s^T \mathbf{a}(\theta) \|\dot{\mathbf{a}}(\theta)\|^2 - \frac{M |\mathbf{a}^H(\theta) \mathbf{R}_s^T \dot{\mathbf{a}}(\theta)|^2}{\mathbf{a}^H(\theta) \mathbf{R}_s^T \mathbf{a}(\theta)} \right)} \quad (67)$$

APPENDIX C

FISHER INFORMATION MATRIX DERIVATION

The FIM [9] for estimating the vector $\boldsymbol{\xi} = [\theta^T \tilde{\alpha}^T]^T$ from the sufficient statistic $\tilde{\mathbf{y}}$, according to the model in (7), is given by

$$[\mathbf{J}(\boldsymbol{\xi})]_{l,p} = \frac{2}{\sigma_w^2} \text{Re} \left\{ \frac{\partial \boldsymbol{\mu}^H}{\partial \xi_l} \frac{\partial \boldsymbol{\mu}}{\partial \xi_p} \right\} \quad l, p = 1, \dots, 3L \quad (57)$$

where $\boldsymbol{\mu} \triangleq \sum_{l=1}^L \alpha_l \mathbf{d}_\beta(\theta_l)$.

First we calculate the partial derivations

$$\frac{\partial \boldsymbol{\mu}}{\partial \theta_l} = \alpha_l \dot{\mathbf{d}}_\beta(\theta_l) \quad (58)$$

$$\frac{\partial \boldsymbol{\mu}}{\partial \tilde{\alpha}_l} = (1, j) \otimes \mathbf{d}_\beta(\theta_l) \quad (59)$$

where $\dot{\mathbf{d}}_\beta(\theta) \triangleq \partial \mathbf{d}_\beta(\theta) / \partial \theta$ and $\otimes$ stands for the Kronecker product. The elements of FIM (57) are given by

$$\begin{aligned} J_{\theta_l \theta_p} &= \frac{2}{\sigma_w^2} \text{Re} \left\{ \alpha_l^* \dot{\mathbf{d}}_\beta^H(\theta_l) \alpha_p \dot{\mathbf{d}}_\beta(\theta_p) \right\} \\ &= \frac{2}{\sigma_w^2} \text{Re} \left\{ \alpha_l^* \alpha_p \text{tr} \left(\dot{\mathbf{d}}_\beta^H(\theta_l) \dot{\mathbf{d}}_\beta(\theta_p) \right) \right\} \\ &= \frac{2N}{\sigma_w^2} \text{Re} \left\{ \alpha_l^* \alpha_p \text{tr} \left(\text{vec} \left(\dot{\mathbf{A}}(\theta_l) \mathbf{U} \mathbf{\Lambda}^{1/2} \right)^H \right. \right. \\ &\quad \left. \left. \times \text{vec} \left(\dot{\mathbf{A}}(\theta_p) \mathbf{U} \mathbf{\Lambda}^{1/2} \right) \right) \right\} \\ &= \frac{2N}{\sigma_w^2} \text{Re} \left\{ \alpha_l^* \alpha_p \text{tr} \left(\mathbf{\Lambda}^{1/2} \mathbf{U}^H \dot{\mathbf{A}}^H(\theta_l) \dot{\mathbf{A}}(\theta_p) \mathbf{U} \mathbf{\Lambda}^{1/2} \right) \right\} \\ &= \frac{2N}{\sigma_w^2} \text{Re} \left\{ \alpha_l^* \alpha_p \text{tr} \left(\dot{\mathbf{A}}(\theta_p) \mathbf{R}_s \dot{\mathbf{A}}^H(\theta_l) \right) \right\} \end{aligned} \quad (60)$$

$$\begin{aligned} \mathbf{J}_{\tilde{\alpha}_l \tilde{\alpha}_p} &= \frac{2N}{\sigma_w^2} \text{Re} \left\{ ((1, j) \otimes \mathbf{d}_\beta(\theta_l))^H ((1, j) \otimes \mathbf{d}_\beta(\theta_p)) \right\} \\ &= \frac{2N}{\sigma_w^2} \text{Re} \left\{ (1, j)^H (1, j) \left(\text{vec} \left(\dot{\mathbf{A}}(\theta_l) \mathbf{U} \mathbf{\Lambda}^{1/2} \right) \right)^H \right. \\ &\quad \left. \times \text{vec} \left(\dot{\mathbf{A}}(\theta_p) \mathbf{U} \mathbf{\Lambda}^{1/2} \right) \right\} \\ &= \frac{2N}{\sigma_w^2} \text{Re} \left\{ (1, j)^H (1, j) \text{tr} \left(\mathbf{A}(\theta_p) \mathbf{R}_s \mathbf{A}^H(\theta_l) \right) \right\} \end{aligned} \quad (61)$$

$$\begin{aligned} J_{\theta_l \tilde{\alpha}_p} &= \frac{2}{\sigma_w^2} \text{Re} \left\{ \alpha_l^* \dot{\mathbf{d}}_\beta^H(\theta_l) ((1, j) \otimes \mathbf{d}_\beta(\theta_p)) \right\} \\ &= \frac{2N}{\sigma_w^2} \text{Re} \left\{ \alpha_l^* \text{tr} \left(\mathbf{A}(\theta_p) \mathbf{R}_s \dot{\mathbf{A}}^H(\theta_l) \right) (1, j) \right\} \end{aligned} \quad (62)$$

where $\dot{\mathbf{A}}(\theta)$ is the derivation of $\mathbf{A}(\theta)$ with respect to θ . In a single target scenario, the CRB on DOA estimation can be obtained by inserting (60)–(62) as shown in (63) at the top of the page, where $\text{SNR} = N|\alpha|^2/\sigma_w^2$. By substitution of (5) into (63), the elements of the FIM can be expressed as

$$\begin{aligned} J_{\theta\theta} &= \frac{2N|\alpha|^2}{\sigma_w^2} \text{tr} \left(\dot{\mathbf{A}}(\theta) \mathbf{R}_s \dot{\mathbf{A}}^H(\theta) \right) \\ &= \frac{2N|\alpha|^2}{\sigma_w^2} \text{tr} \left([\dot{\mathbf{a}}(\theta) \mathbf{a}^T(\theta) + \mathbf{a}(\theta) \dot{\mathbf{a}}^T(\theta)] \right. \\ &\quad \left. \times \mathbf{R}_s [\mathbf{a}^*(\theta) \mathbf{a}^H(\theta) + \mathbf{a}^*(\theta) \dot{\mathbf{a}}^H(\theta)] \right) \\ &= \frac{2N|\alpha|^2}{\sigma_w^2} \left(\mathbf{a}^T(\theta) \mathbf{R}_s \mathbf{a}^*(\theta) \dot{\mathbf{a}}^H(\theta) \dot{\mathbf{a}}(\theta) \right. \\ &\quad \left. + M \dot{\mathbf{a}}^T(\theta) \mathbf{R}_s \dot{\mathbf{a}}^*(\theta) \right) \\ &= \frac{2N|\alpha|^2}{\sigma_w^2} \mathbf{a}^H(\theta) \mathbf{R}_s^T \mathbf{a}(\theta) (\|\dot{\mathbf{a}}(\theta)\|^2 + M) \end{aligned} \quad (64)$$

$$\begin{aligned} \mathbf{J}_{\tilde{\alpha}\tilde{\alpha}} &= \frac{2N}{\sigma_w^2} \text{Re} \left\{ (1, j)^H (1, j) \text{tr} \left(\mathbf{A}(\theta) \mathbf{R}_s \mathbf{A}^H(\theta) \right) \right\} \\ &= \frac{2N}{\sigma_w^2} M \mathbf{a}^T(\theta) \mathbf{R}_s \mathbf{a}^*(\theta) \mathbf{I}_2 \\ &= \frac{2N}{\sigma_w^2} M \mathbf{a}^H(\theta) \mathbf{R}_s^T \mathbf{a}(\theta) \mathbf{I}_2 \end{aligned} \quad (65)$$

$$\begin{aligned} \mathbf{J}_{\theta\tilde{\alpha}} &= \frac{2N}{\sigma_w^2} \text{Re} \left\{ \alpha^* \text{tr} \left(\mathbf{A}(\theta_p) \mathbf{R}_s \dot{\mathbf{A}}^H(\theta_l) \right) (1, j) \right\} \\ &= \frac{2NM}{\sigma_w^2} \text{Re} \left\{ \alpha^* \mathbf{a}^T(\theta) \mathbf{R}_s \dot{\mathbf{a}}^*(\theta) (1, j) \right\} \\ &= \frac{2NM}{\sigma_w^2} \text{Re} \left\{ \alpha^* \mathbf{a}^H(\theta) \mathbf{R}_s^T \dot{\mathbf{a}}(\theta) (1, j) \right\} \end{aligned} \quad (66)$$

where it is assumed that the equality $\dot{\mathbf{a}}^H(\theta) \mathbf{a}(\theta) = 0$ is satisfied if the array origin is at the array centroid, i.e., $\sum_{m=1}^M \mathbf{x}_m = 0$. The matrix $\mathbf{I}_2$ denotes an identity matrix of size two. Finally, the CRB on DOA for a single target is given by (67) as shown at the top of the page.

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